Md Mumtahin Habib Ullah Mazumder

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Website | Google Scholar | GitHub | LinkedIn

Research Profile

Ph.D. researcher at the intersection of **learning theory, algorithms, optimization, and empirical deep learning**. Research includes provable algorithms under stochastic feedback, probabilistic graph optimization, physics-informed data augmentation, and neural-network training for wireless systems. Interested in theoretical and experimental questions about **learning dynamics and adaptation in deep networks**.

Education

University of Utah, Salt Lake City, UT Aug 2023 – Dec 2027 (expected)
Ph.D. in Computer Science | Advisor: Prof. Aditya Bhaskara

University of Utah, Salt Lake City, UT Aug 2023 – Apr 2026
M.S. in Computer Science | GPA: 3.98/4.00

Islamic University of Technology, Bangladesh Jan 2017 – Mar 2021
B.Sc. in Computer Science and Engineering | GPA: 3.88/4.00

Relevant coursework: Theory of Machine Learning, Convex Optimization, Linear Algebra, Approximation/Randomized Algorithms, High-Dimensional Data Analysis, Deep Learning with NLP.

Research Experience

Graduate Research Assistant | University of Utah Aug 2023 – Present
Learning theory, graph algorithms, and neural methods for wireless systems

- **Stochastic subset selection.** Developing and analyzing greedy/local-search algorithms for selecting high-utility subsets from stochastic feasibility feedback, including approximation and oracle/query-complexity guarantees and sample-efficient learning.
- **Probabilistic graph optimization.** Co-developed algorithms for edge augmentation in probabilistic graphs, studying approximation guarantees and hardness results for improving information reach under randomized propagation.
- **Neural learning and generalization.** Developed physics-informed received-signal-strength (RSS) data augmentation for wireless localization; evaluated learned models under sparse training data, distribution shift, and multi-transmitter settings (CoNEXT 2025).

Research Assistant | IDS and Imbalanced Learning Research Nov 2020 – Dec 2022
Algorithm design and empirical machine-learning evaluation

- Developed feature extraction/reduction methods for intrusion detection and contributed to an evidential dynamic-neighborhood method for learning from imbalanced data; evaluated methods in classification experiments and co-authored peer-reviewed publications.

Selected Publications & Preprints

Local Search for Subset Selection with Stochastic Feasibility Feedback. A. Bhaskara, S. K. Kasera, **M. M. H. U. Mazumder**. *Manuscript submitted to ACM SIGMETRICS 2027.* [Manuscript]

Bridging Data Gaps: Enhancing Wireless Localization with Physics-Informed Data Augmentation. M. M. H. U. Mazumder, F. B. Mitchell, A. Bhaskara, S. K. Kasera, N. Patwari. *Proc. ACM on Networking (CoNEXT 2025).* [Paper].

Optimizing Probabilistic Propagation in Graphs by Adding Edges. A. Bhaskara, A. Crane, S. Jain, **M. M. H. U. Mazumder**, B. D. Sullivan, P. Yalamanchili. *Preprint, under review.* [arXiv].

An Evidential Dynamic Neighborhood Approach to Learn from Imbalanced Data. M. E. Kadir, **M. M. H. U. Mazumder**, M. M. Alam, M. Shoyaib. *Annals of Data Science*, 2026. [Paper].

Selected Deep Learning & Research Software Projects

5G RAN Scheduler Simulator and Deep Reinforcement Learning Testbed [GitHub]
Python, PyTorch, PPO, NVIDIA Sionna, CUDA, Slurm

- Implemented actor–critic RL training and evaluation for multi-cell radio-resource scheduling, including PPO, generalized advantage estimation, action masking, expert-guided training, and multicell rollouts; coupled policies to a physics-based network simulator.

U-Net RF Transmitter Localization [GitHub]
Python, PyTorch, U-Net, CNNs

- Built and experimentally evaluated neural spatial-heatmap prediction from sparse RSS measurements across eight wireless datasets; implemented elevation-aware inputs, sensor calibration, custom training losses, and HPC-based evaluation.

Primal-Dual Automatic Question Generation [GitHub]
Python, PyTorch, BART, Hugging Face

- Implemented coupled question-generation and answer-extraction objectives for a BART-based model, with ablation experiments and evaluation using BLEU/ROUGE and QA-consistency metrics on SQuAD and HotpotQA.

Additional Experience & Academic Service

Research Intern, Samsung R&D Institute Bangladesh Nov 2019 – Jan 2020
Developed a deep-learning proof of concept for S-Pen interaction and associated data-processing experiments.

Machine Learning Instructor, University of Utah REU May – Aug 2024
Delivered machine-learning lectures and mentored undergraduate research projects.

Lecturer, United International University Jul 2021 – Jul 2023
Taught undergraduate computer-science courses and supervised student projects.

Academic service: Reviewer, SODA 2026 and UAI 2026; Shadow PC Member, SIGMETRICS 2026.

Technical Skills

Core: Python, PyTorch, machine learning, reinforcement learning (PPO), neural-network experimentation, algorithms, mathematical optimization.

Additional: C/C++, Java, SQL, Git, Linux, CUDA, HPC/Slurm, scikit-learn, Hugging Face, NumPy, Pandas, Docker, NVIDIA Sionna.